IILM UNIVERSITY | GREATER NOIDA, UTTAR PRADESH

INTERNSHIP PRESENTATION

A I / M L V I R T U A L I N T E R N S H I P

Classification & Prediction Systems using ML

- S M A R T E D I N N O V A T I O N S

Bappaditya Das

25SCS1003002304

B.Tech in CSE with AI/ML

Try Pitch

School of Computer Science and Engineering

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INTRODUCTION

This report documents the work undertaken during a Virtual Internship in Artificial Intelligence and Machine Learning, completed with SmartED Innovations in association with IILM University, Greater Noida, as part of the academic requirement for the 3rd semester B.Tech CSE curriculum.

The internship combined theoretical instruction — delivered across six structured modules — with hands-on, project-based learning, culminating in two independent Machine Learning systems.

PROJECT 1:
NYC ROOM CLASSIFICATION

Categorising NYC room/listing data using supervised classification algorithms.

PROJECT 2:
MENTAL HEALTH SCORE PREDICTION

Estimating a wellness score from lifestyle & behavioural indicators using regression.

CANDIDATE'S PROFILE

BAPPADITYA DAS

25SCS1003002304

IILM UNIVERSITY

ORG: SMARTED INNOVATIONS

6 WEEKS

About the Organization

SmartED Innovations

BENGALURU-BASED ED-TECH ORGANIZATION

01

ACADEMIC PARTNERSHIPS

Partners with universities to deliver industry-oriented virtual internship programs across emerging technology domains.

02

CORE DOMAINS

Focus areas include Artificial Intelligence, Data Science, and Web Development, Cyber Security.

03

HYBRID LEARNING

Programs combine conceptual/theory sessions with guided, hands-on project work.

04

PRACTICAL EXPERIENCE

Enables students to apply classroom learning to practical problems while working remotely.

Problem Statement

CLASSIFICATION

NYC ROOM CLASSIFICATION

Classify the Given Dataset & Draw insights and classify them in categories

REGRESSION

MENTAL HEALTH SCORE PREDICTION

Build a predictive model capable of estimating a Mental Health Score for an individual, and identify key contributing factors.

OBJECTIVES & SCOPE

1

Understand and apply the end-to-end Machine Learning workflow, from raw data to a working predictive model.

2

Perform data cleaning, exploratory data analysis (EDA), and feature engineering on real-world style datasets.

3

Evaluate model performance using appropriate metrics and visualise results for interpretability.

4

Gain hands-on exposure to Python's data-science ecosystem (Pandas, NumPy, Scikit-learn, Matplotlib/Seaborn).

5

Understanding and correctly applying the ML pipeline, rather than achieving production-grade accuracy.

SYSTEM ARCHITECTURE

Data Collection → Data Preprocessing → Feature Engineering → Model Training → Model Evaluation → Prediction Output

Technology and Tools :

01	**LANGUAGE**	PYTHON 3
02	**DATA HANDLING**	PANDAS, NUMPY, DATAFRAMES, EDA
03	**MACHINE LEARNING**	SCIKIT-LEARN
04	**Visualization**	MATPLOTLIB, SEABORN
05	**Environment**	JUPYTER NOTEBOOK/GOOGLE COLAB

Methodology

NYC ROOM CLASSIFICATION

1

DATA LOADING & CLEANING

Loaded with Pandas; missing values handled via imputation/row removal; high-cardinality ID columns dropped.

2

EXPLORATORY DATA ANALYSIS

Distribution of listings by borough, room type, and price range visualised before modelling.

3

FEATURE ENCODING

Categorical features (neighbourhood group, room type) encoded via Label/One-Hot Encoding.

4

TRAIN-TEST SPLIT

Dataset split (typically 80:20) using Scikit-learn's train_test_split.

5

MODEL TRAINING

While explaining your thoughts, make sure your tone and writing style stay consistent. Are you aiming to be serious or funny, emotional or factual?

6

EVALUATION

Assessed using accuracy score, confusion matrix, and precision/recall.

Methodology

MENTAL HEALTH SCORE PREDICTION

1

DATA LOADING & CLEANING

Dataset of lifestyle/behavioural indicators (sleep, screen time, exercise, social interaction) loaded and cleaned.

2

EXPLORATORY DATA ANALYSIS

Analyzed variable distributions, correlation heatmaps, and scatter plots of lifestyle metrics-sleep, screen time, and exercise-against the target score to uncover key behavioral trends and identify outliers.

3

FEATURE ENCODING

Numerical features scaled/normalised where required for comparability across ranges.

4

TRAIN-TEST SPLIT

Data split into training and testing sets for unbiased evaluation.

5

MODEL TRAINING

Linear Regression and Random Forest Regressor trained to predict a continuous Mental Health Score.

6

EVALUATION

Assessed using MAE, MSE, and R^2 score; feature importance analysed for influential factors.

EXPECTED OUTCOMES:

1 .

A working classification model that can categorise NYC room listings with reasonable accuracy.

2 .

A working regression model that estimates a Mental Health Score from lifestyle indicators, with insight into influential factors.

3 .

Practical, transferable familiarity with the full ML workflow — data cleaning, EDA, feature engineering, training, and evaluation.

4 .

A foundation for exploring more advanced algorithms and larger, real-world datasets in future coursework.

TRAINING LOG-BOOK

Content of Modules 1-6:

MODULE 1:

Intro To AI & Python Essentials:

1. What is AI? Turing Test to today
2. Types of AI
3. Python essentials
4. NumPy & Pandas
5. Math foundations & statistics

MODULE 2:

Problem Solving Agents & Search:

1. Agents & rationality
2. PEAS framework
3. Problem formulation & state space
4. Uninformed search
5. Informed search

MODULE 3:

Rule-Based Systems & Traditional ML:

1. Rule-based & expert systems
2. Supervised learning: regression & classification
3. Evaluation metrics
4. Bias-variance tradeoff
5. End-to-end ML pipeline

MODULE 4:

Unsupervised Learning & Probabilistic Reasoning:

1. K-Means & hierarchical clustering
2. Dimensionality reduction
3. Bayes' theorem
4. Naive Bayes classifier
5. Bayesian networks

MODULE 5:

Deep Learning & Computer Vision:

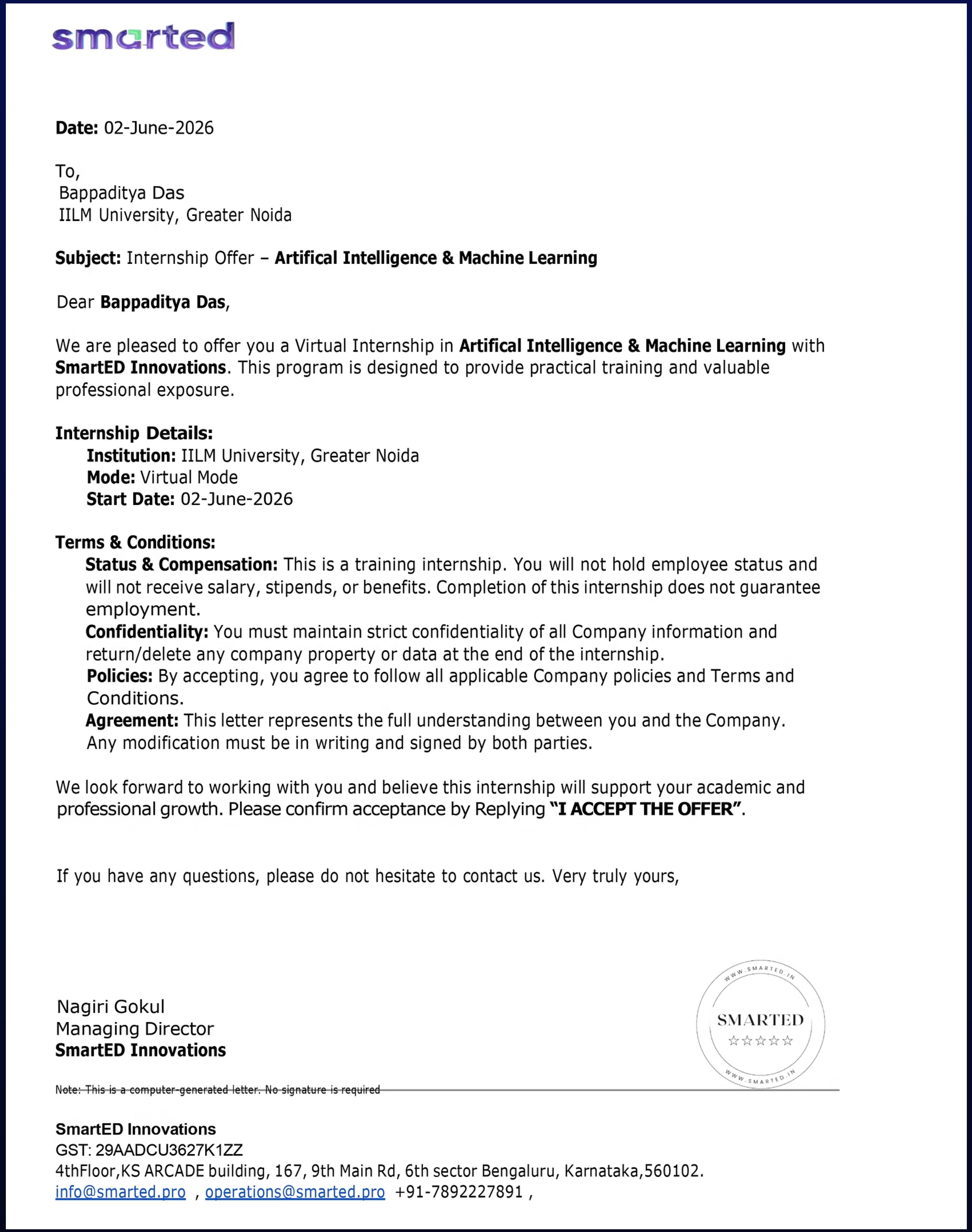
1. Neural network foundations
2. Backpropagation & training
3. TensorFlow, Keras, PyTorch

MODULE 6:

NLP, Generative AI & Industrial Applications:

1. Text preprocessing
2. Vectorisation techniques
3. Sentiment analysis pipeline
4. Transformers, GPT, prompt engineering

CERTIFICATIONS



smarted

Date: 02-June-2026

To,
Bappaditya Das
IILM University, Greater Noida

Subject: Internship Offer – **Artifical Intelligence & Machine Learning**

Dear **Bappaditya Das**,

We are pleased to offer you a Virtual Internship in **Artifical Intelligence & Machine Learning** with **SmartED Innovations**. This program is designed to provide practical training and valuable professional exposure.

Internship Details:

Institution: IILM University, Greater Noida
Mode: Virtual Mode
Start Date: 02-June-2026

Terms & Conditions:

Status & Compensation: This is a training internship. You will not hold employee status and will not receive salary, stipends, or benefits. Completion of this internship does not guarantee employment.
Confidentiality: You must maintain strict confidentiality of all Company information and return/delete any company property or data at the end of the internship.
Policies: By accepting, you agree to follow all applicable Company policies and Terms and Conditions.
Agreement: This letter represents the full understanding between you and the Company. Any modification must be in writing and signed by both parties.

We look forward to working with you and believe this internship will support your academic and professional growth. Please confirm acceptance by Replying **"I ACCEPT THE OFFER"**.

If you have any questions, please do not hesitate to contact us. Very truly yours,

Nagiri Gokul
Managing Director
SmartED Innovations

SMARTED

Note: This is a computer-generated letter. No signature is required

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Internship Offer Letter and Certificate of Completion issued by SmartED Innovations.